VANISHING OF THE LOCAL COHOMOLOGY OF THE PUNCTURED CLOSED SET MAKES THE COMPARISON MAP BETWEEN TWO LOCAL COHOMOLOGY MODULES SURJECTIVE, INJECTIVE OR BIJECTIVE

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Allowed standard prerequisites (editorial).

- Local cohomology H^i_{V(I)}(M) of a module with support in the closed subset V(I), and its localisation formula H^i_{V(I)}(M)_\mathfrak p = H^i_{\mathfrak pA_\mathfrak p}(M_\mathfrak p). The author's section on generalities about local cohomology, in which the counted statement occurs. Supplies the two modules compared by the counted map and the local vanishing hypothesis.
- The lemma that elements of a local cohomology module come from a strictly bigger finitely generated ideal. Author lemma lemma-elements-come-from-bigger, whose statement is reproduced verbatim as a context span of this card (source lines 451-470). Provides the step that increases the auxiliary ideal in both halves of the proof and thereby contradicts maximality.

Lemma 0.1. *Let $I \subset I' \subset A$ be finitely generated ideals of a Noetherian ring A. Let M be an A-module. Let $i \geq 0$ be an integer. Consider the map*

$$\Psi : H^i_{V(I')}(M) \to H^i_{V(I)}(M)$$

The following are true:

(1) *if $H^i_{\mathfrak p A_\mathfrak p}(M_\mathfrak p) = 0$ for all $\mathfrak p \in V(I) \setminus V(I')$, then Ψ is surjective,*
(2) *if $H^{i-1}_{\mathfrak p A_\mathfrak p}(M_\mathfrak p) = 0$ for all $\mathfrak p \in V(I) \setminus V(I')$, then Ψ is injective,*
(3) *if $H^i_{\mathfrak p A_\mathfrak p}(M_\mathfrak p) = H^{i-1}_{\mathfrak p A_\mathfrak p}(M_\mathfrak p) = 0$ for all $\mathfrak p \in V(I) \setminus V(I')$, then Ψ is an isomorphism.*

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Lemma 0.2. *Let $I \subset A$ be a finitely generated ideal of a ring A. Let $\mathfrak{p}$ be a prime ideal. Let M be an A-module. Let $i \geq 0$ be an integer and consider the map*

$$\Psi : \operatorname{colim}_{f \in A, f \notin \mathfrak{p}} H^i_{V((I,f))}(M) \longrightarrow H^i_{V(I)}(M)$$

Then

(1) $\operatorname{Im}(\Psi)$ *is the set of elements which map to zero in $H^i_{V(I)}(M)_\mathfrak{p}$,*
(2) *if $H^{i-1}_{V(I)}(M)_\mathfrak{p} = 0$, then Ψ is injective,*
(3) *if $H^{i-1}_{V(I)}(M)_\mathfrak{p} = H^i_{V(I)}(M)_\mathfrak{p} = 0$, then Ψ is an isomorphism.*

Proof. Proof of (1). Let $\xi \in H^i_{V(I)}(M)$. Since A is Noetherian, there exists a largest ideal $I \subset I'' \subset I'$ such that ξ is the image of some $\xi'' \in H^i_{V(I'')}(M)$. If $V(I'') = V(I')$, then we are done. If not, choose a generic point $\mathfrak{p} \in V(I'')$ not in $V(I')$. Then we have $H^i_{V(I'')}(M)_\mathfrak{p} = H^i_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p}) = 0$ by assumption. By Lemma 0.2 we can increase I'' which contradicts maximality.

Proof of (2). Let $\xi' \in H^i_{V(I')}(M)$ be in the kernel of Ψ. Since A is Noetherian, there exists a largest ideal $I \subset I'' \subset I'$ such that ξ' maps to zero in $H^i_{V(I'')}(M)$. If $V(I'') = V(I')$, then we are done. If not, then choose a generic point $\mathfrak{p} \in V(I'')$ not in $V(I')$. Then we have $H^{i-1}_{V(I'')}(M)_\mathfrak{p} = H^{i-1}_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p}) = 0$ by assumption. By Lemma 0.2 we can increase I'' which contradicts maximality.

Part (3) is formal from parts (1) and (2). □